Course Code	21CSC569J	Course Name	Fundamentals of Artificial Intelligence	Course Category	C	Professional Core Course	L	T	P	C
							3	0	2	4

Pre-requisite Courses	*Nil*	Co-requisite Courses	*Nil*	Progressive Courses	*Nil*
Course Offering Department	Computational Intelligence	Data Book / Codes/Standards	*Nil*		

Course Learning Rationale (CLR)	The purpose of learning this course is to:
CLR-1 :	Understand the basic data structures involved in python to perform exploratory data analysis
CLR-2 :	Understands data analysis and visualization using python
CLR-3 :	Learn the distinction between optimal reasoning Vs human like reasoning and formulate an efficient problem space for a problem expressed in natural language. Also select a search algorithm for a problem and estimate its time and space complexities.
CLR-4 :	Understand the concepts of constraint satisfaction problems and optimal decision making
CLR-5 :	Understand the various methods of Knowledge representation and Reasoning with examples.

Course Outcomes (CO):	At the end of this course, learners will be able to:	Program Outcomes (PO) PO1	PO2	PO3
CO-1:	Do the exploratory data analysis techniques in Python environment.	2	2	2
CO-2:	Formulate and use appropriate data analysis and visualization techniques for their data	2	2	2
CO-3:	Identify and design the suitable AI Agents and also Apply the basic search techniques for problem solving	3	2	2
CO-4:	Solve the constraint satisfaction problems with optimal decision.	2	3	3
CO-5:	Solve the AI problems by incorporating the optimal knowledge Representation and Reasoning techniques	2	2	3

Unit-1 Introduction to Python for Data Science (12 hrs)

Python Programming Introduction, Python Data structures and Functions, Basic Python programs, Introduction to Data science , Understanding the nature of Data -Types of Data , Introduction to libraries: NumPy, Pandas, Matplotlib , Data types in Python relevant to data science ,Applications using Python libraries, Hands-on practice with NumPy and Pandas.

Lab1 : Python Basics and Control Structures
Lab2 : Create a NumPy array and perform basic operations
Lab3: Use Pandas to load a CSV file, display the first 10 rows, and perform basic data manipulation (sorting, filtering).

Unit-2 Data Analysis and Visualization with Python (12 hrs) Data Loading and Cleaning: Reading and Writing Data with Pandas, Data Manipulation with Pandas, Exploratory Data Analysis (EDA): Descriptive Statistics, Data Visualization with Matplotlib and Seaborn , Correlation and Covariance , Advanced Data Visualization- Interactive Plots with Plotly. **Lab 4 :** Load a dataset into a Pandas DataFrame and clean the data (handle missing values, drop unnecessary columns) and Perform group by operations and calculate summary statistics. **Lab 5 :** Create a variety of plots using Matplotlib, Use Seaborn to create advanced visualizations (box plot, heatmap) and customize them (titles, labels, colors).
Unit-3 Introduction to AI and Problem Solving/ Searching Techniques (12 hrs) Introduction to AI: Problems of AI, AI techniques, Intelligent Agents and Environment, Defining the problem as state space search, production system, problem characteristics, and issues in the design of search programs. Problem solving agents, searching for solutions; uniform search strategies: breadth first search, depth first search, depth limited search, bidirectional search, comparing uniform search strategies. Heuristic search strategies Greedy best -first search, A* search, AO* search, memory bounded heuristic search: local search algorithms & optimization problems: Hill climbing search, simulated annealing search, local beam search. **Lab 6:** Developing agent programs for real world problems **Lab 7:** Implementation and Analysis solution for TSP **Lab 8:** Developing Best first search and A*Algorithm for real world problems **Lab 9:**Implementation of Simulated Annealing Algorithm and Hill Climbing for real world problem
Unit-4 Constraint Satisfaction Problems and Game Theory: (12 hrs) Local search for constraint satisfaction problems. Adversarial search, Games, optimal decisions & strategies in games, the minimax search procedure, alpha-beta pruning, additional refinements, iterative deepening. **Lab 10:** Implementation of constraint satisfaction problems **Lab 11:** Implementation of alpha-beta pruning algorithm for an application **Lab 12:** Develop an agent using Minimax algorithm to play Tic-Tac-Toe:
Unit-5 Knowledge Representation and Reasoning for various AI applications (12 hrs) Statistical Reasoning: Probability and Bayes' Theorem, Certainty Factors and Rule-Base Systems, Bayesian Networks, Dempster-Shafer Theory, Fuzzy Logic. AI for knowledge representation, rule-based knowledge representation, procedural and declarative knowledge, Logic programming, Forward and backward reasoning. Applications: Expert Systems, Decision Support System, Generative AI in Natural Language Processing (NLP) AI-based Programming tools. **Lab13:** Implementation of unification and resolution for real world problems. **Lab14** Implementation of knowledge representation schemes - use cases **Lab 15:** Develop Chatbot for SRM Admission Enquiry

Learning Resources	1. Joel Grus," Data Science from Scratch: First Principles with Python, Second Edition.O'Reilly,2019 2. Charles R. Severance , "Python for Everybody Exploring Data Using Python", Charles Severance, 2016.	https://towardsai.net/p/machine-learning/large-language-models-and-gpt-4-architecture-and-openai-api

	3. "Artificial Intelligence" E. Rich and K. Knight, Mc Graw Hill Publishers INC, 3rd Edition 2017. 4. "A First Course in Artificial Intelligence", Deepak Khemani, McGraw Hill Education, 2013. 5. Gerhard Weiss, —Multi Agent Systems‖, Second Edition, MIT Press, 2013	

	Bloom's Level of Thinking	**Continuous Learning Assessment (CLA) - By the Course Faculty**				**By The CoE**	
		Formative CLA-I Average of unit test (45%)		**Life Long* Learning** CLA-II- Practice (15%)		**Summative** Final Examination (40% weightage)	
		Theory	Practice	Theory	Practice	Theory	Practice
Level 1	Remember	25%	-	-	15%	15%	-
Level 2	Understand	25%	-	-	25%	20%	-
Level 3	Apply	20%	-	-	20%	15%	-
Level 4	Analyze	15%	-	-	20%	25%	-
Level 5	Evaluate	15%	-	-	20%	25%	-
Level 6	Create	-	-	-	-	-	-
	Total	100 %		100 %		100 %	

Experts from Industry	Experts from Higher Technical Institutions	Internal Experts
Mr.JagatheeswaranSenthilvelan, Managing Partner , ProtoHubs.io		1. Dr.M. Karpagam, SRMIST 2. Dr.C. Lakshmi, SRMIST